THE NOVA RESEARCH CURRICULUM

# Computer Science and skills

Grade 7 · Term 1

Student edition

Outcomes: CS-7.1 (part) Python loops and lists; RO-7.1; FL-7.1; RP-7.1; AR; PE.

## Lesson notes

- **CS-7.1 (Weeks 1–12):** `for i in range(5):` repeats a block; `while` repeats while a condition is true (always change the variable to avoid an endless loop). Lists: `marks = [70, 85, 90]`, `marks.append(60)`, `len(marks)`, `marks[0]`; loop through a list to find the total and average. Browser-based Python editor; no installation needed.
- **RO-7.1:** in a free online microcontroller simulator (for example Wokwi or Tinkercad Circuits), blink an LED, read a button, then turn on an LED when a light sensor reads dark. Real kits later from the partner lab.
- **FL-7.1:** profit and loss in a small stall; Zakat calculation (links to MA-7.2).
- **RP-7.1:** choose a survey question (for example "How do students travel to school?"), design 5 questions, collect data from 20 people with permission and no names.
- **AR, PE:** sketching from life and perspective; football tactics, fitness test, first-aid basics (recovery position, cleaning a cut, when to call an adult).

## Workbook pages

| Page | Task |
|---|---|
| 1–2 | Trace five loop programs; predict the output. |
| 3 | Marks list program: total, average, highest. |
| 4 | Simulator circuit log. |
| 5 | Survey plan and data table. |

## Question bank

| # | Outcome | Question | Your answer |
|---|---|---|---|
| 1 | CS-7.1 | How many times does `for i in range(4):` repeat? | 3 / 4 / 5 / forever |
| 2 | CS-7.1 | What is printed by `print(list(range(3)))`? | [1, 2, 3] / [0, 1, 2] / [0, 1, 2, 3] / 3 |
| 3 | CS-7.1 | `marks = [70, 85, 90]`. What is `marks[1]`? | 70 / 85 / 90 / an error |
| 4 | CS-7.1 | Which adds 60 to the end of the list? | marks.add(60) / marks.append(60) / marks + 60 / marks.insert() |

| # | Outcome | Question | Your answer |
|---|---|---|---|
| 5 | CS-7.1 | A while loop whose condition never becomes false runs forever. True or false. | True / False |
| 6 | CS-7.1 | Write a loop that prints the numbers 1 to 5. | _____ |
| 7 | RO-7.1 | A light sensor is an … device. | output / input / power / storage |
| 8 | RP-7.1 | Survey answers should be collected without names to protect privacy. True or false. | True / False |